Qassim University جامعة القصيم

Minimizing a Nonlinear Objective Function Under a Single Linear Constraint Using Lagrange Multipliers

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CS 348 - COURSE PROJECT

OBJECTIVE FUNCTION IMPLEMENTATION AND ANALYSIS

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1.Introduction

Optimization is a central subject in applied mathematics, where one looks for the best possible outcome under given limitations. Though many optimization problems indeed involve maximization of some performance or profit, minimization problems are no less relevant: quite often, the objective in view is minimization of cost or error or energy consumption.

In this project, we consider a simple constrained minimization problem with two decision variables and one linear resource constraint. We apply the method of Lagrange Multipliers to obtain an analytical solution for a nonlinear objective function in order to find the minimum point corresponding to the given constraint. The approach allows us to connect the underlying mathematical concepts with implementation and visualization in a computational setting.

2. Problem Statement

We consider a system with two variables, x and y, representing quantities that must satisfy a fixed resource limitation. The goal is to **minimize** a nonlinear objective function under a linear constraint.

- **Decision variables:** x and y
- **Objective:** Minimize the quadratic function C(x, y)
- **Constraint:** Linear equality representing limited resources

This setup allows us to determine the minimum feasible value of the objective function analytically and numerically.

3. Mathematical Formulation

a) Objective Function

We define a convex quadratic function:

$C(x,y)=3x^2+4y^2$

This ensures a well-defined minimum.

b) Constraint

The variables must satisfy a linear resource constraint (equality constraint) and non-negativity constraints (boundary constraints) :

$x+y=80$

$x \geq 0, y \geq 0$

c) Lagrangian Function

To solve the constrained minimization, we construct the Lagrangian:

$L(x,y,\lambda)=3x^2+4y^2+\lambda(80-x-y)$

d) Optimality Conditions

We take the partial derivatives and set them to zero:

$\partial L/\partial x=6x-\lambda=0$

$\partial L/\partial y=8y-\lambda=0$

$\partial L/\partial \lambda=80-x-y=0$

Solving this system gives the point (x^{*},y^{*}) that minimizes $C(x,y)$ while satisfying the constraint.

4. Lagrange Method

This part is the solution to a constrained optimization problem involving a quadratic cost function and one linear constraint. The method of Lagrange multipliers is applied to determine the stationary point, followed by a boundary evaluation to confirm the global minimum within the nonnegative domain.

Problem Statement

- Decision variables: (x, y)
- Objective function: C(x,y) = 3x^2 + 4y^2
- Constraint: x + y = 80
- Domain restriction: x >= 0, y >= 0
- Goal: Determine the values of x and y that minimize the cost function while satisfying the constraint.

<u>Lagrangian Formulation and First-Order Conditions</u>

Lagrangian Function : L(x,y,λ) = 3x^2 + 4y^2 + λ(80 - x - y)

Partial Derivatives

- With respect to x:

∂L/∂x = 6x - λ = 0

⇒ x = λ / 6

- With respect to y:

∂L/∂y = 8y - λ = 0

⇒ y = λ / 8

- With respect to λ:

$\partial L/\partial\lambda = 80 - x - y = 0$

Solving for the Stationary Point

Substitute x = λ / 6 and y = λ / 8 into the constraint:

$\lambda/6 + \lambda/8 = 80$

$\lambda(7/24) = 80$

$\lambda = (80 * 24)/7$

Compute the optimal values:

$x^* = \lambda / 6 = 320 / 7 \approx 45.7143$

$y^* = \lambda / 8 = 240 / 7 \approx 34.2857$

Minimum cost at the stationary point:

$C^* = 3*(320/7)^2 + 4*(240/7)^2$

$= 537600 / 49$

≈ 10971.43

Boundary Evaluation :

The feasible region along the line $x + y = 80$ with nonnegativity consists of two endpoints:

- At (80, 0):

$C = 3*(80)^2 = 19200$

- At (0, 80):

  $C = 4*(80)^2 = 25600$

Comparison:

$C(x^*, y^*) \approx 10971.43 < 19200, 25600$

Thus, the stationary point provides a lower cost than either boundary point, confirming that the minimum occurs in the interior of the feasible region.

Because the coefficient associated with y^2 is larger than the one associated with x^2, the optimal solution assigns a greater share of the total to x. This reduces the overall cost while still satisfying the constraint. At the optimal point, the gradient of the cost function is aligned with the gradient of the constraint, which characterizes the Lagrange multiplier condition.

$x^* = 320 / 7 \approx 45.7143$

$y^* = 240 / 7 \approx 34.2857$

$C_{min} = 537600 / 49 \approx 10971.43$

The Lagrange multiplier method identifies a unique interior minimum, and the boundary check confirms that this point yields the lowest possible cost under the constraint x + y = 80 with nonnegative variables

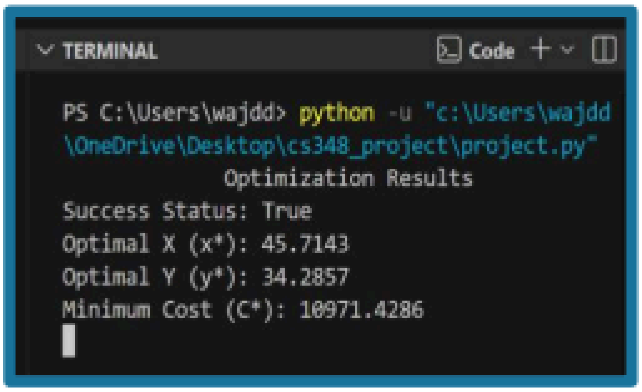
5. Numerical Implementation

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize, Bounds, LinearConstraint

# Objective Function (Minimize C(x, y) = 3x^2 + 4y^2)
def objective_function(X):
    x, y = X
    return 3 * x**2 + 4 * y**2

# Constraint: x + y = 80
# A = [1, 1] , b = 80
linear_constraint = LinearConstraint([1, 1], [80], [80])

# Non-Negativity Bounds: x >= 0, y >= 0
bounds = Bounds([0, 0], [np.inf, np.inf])

initial_guess = [40, 40]

# Run the optimization algorithm
result = minimize(
    objective_function,
    initial_guess,
    method='SLSQP',
    bounds=bounds,
    constraints=linear_constraint
)

# results
x_optimal, y_optimal = result.x
cost_optimal = result.fun


print("             Optimization Results")

print(f"Success Status: {result.success}")
print(f"Optimal X (x*): {x_optimal:.4f}")
print(f"Optimal Y (y*): {y_optimal:.4f}")
print(f"Minimum Cost (C*): {cost_optimal:.4f}")



# Visualization (Contour Plot)

x_vals = np.linspace(0, 80, 100)
y_vals = np.linspace(0, 80, 100)
X, Y = np.meshgrid(x_vals, y_vals)
C = 3 * X**2 + 4 * Y**2


plt.figure(figsize=(10, 7))


levels_to_plot = [5000, 10000, 15000, 19542.86, 25000]
contour = plt.contour(X, Y, C, levels=levels_to_plot, cmap='coolwarm')
```

```python
plt.clabel(contour, inline=True, fontsize=10, fmt='%1.0f')
plt.colorbar(label='Total Cost C(x,y)')

y_constraint = 80 - x_vals
plt.plot(x_vals, y_constraint, 'g--', linewidth=2, label='Production
Constraint: x + y = 80')


plt.plot(x_optimal, y_optimal, 'ko', markersize=8, label='Optimal Point
$(x^*, y^*)$')
plt.annotate(
    f'Optimal:\n({x_optimal:.2f}, {y_optimal:.2f})',
    (x_optimal, y_optimal),
    textcoords="offset points",
    xytext=(10, -10),
    ha='left',
    fontsize=12,
    color='black'
)


plt.title('Cost Minimization using Lagrange (Contour Plot)')
plt.xlabel('Product X Quantity')
plt.ylabel('Product Y Quantity')
plt.legend()
plt.grid(True, linestyle=':', alpha=0.7)
plt.show()
```

6. Description of the Numerical method

The code implements the minimization problem using the Python language and the SciPy library. The SLSQP (Sequential Least Squares Programming) iterative algorithm was utilized via the minimized function to find the optimal allocation that minimizes the quadratic cost function C(x,y) = 3x^2 + 4y^2 ,subject to the equality constraint x + y = 80



Numerical Output and Results:

The following table summarizes the output generated by the Python optimization code, confirming the analytical solution for the **Cost Minimization** problem (C(x,y) = 3x^2 + 4y^2 subject to x+y=80)

Output Variable	Value	Description
Success Status	True	The algorithm successfully found the optimal solution.
Optimal X (x*)	45.7143	The optimal quantity for Product X 320 /7
Optimal Y (y*)	34.2857	The optimal quantity for Product Y 240 /7
Minimum Cost (C*)	10971.43	The calculated minimum cost achieved at the optimal point (537600/49)

7. Visualization

Contour Plot Analysis

The visualization step provides a graphical interpretation of the solution, connecting the mathematical model to the final optimal result. The figure below illustrates the Cost Contours of the objective function $C(x,y) = 3x^2 + 4y^2$ against the production quota constraint $x + y = 80$.

- Contours (Elliptical Lines): These curved lines represent constant levels of Total Cost (C). Since the objective is minimization, the innermost contour lines represent lower cost values.
- Constraint Line (Green Dashed Line): This line represents the feasible region (all combinations of x and y that satisfy $x + y = 80$).
- Optimal Point: The point ($x^*=45.71$, $y^*=34.29$) is the point where the cost function's contour line is tangent (touches but does not cross) the constraint line. This geometric tangency condition confirms the solution derived analytically using the Lagrange method.

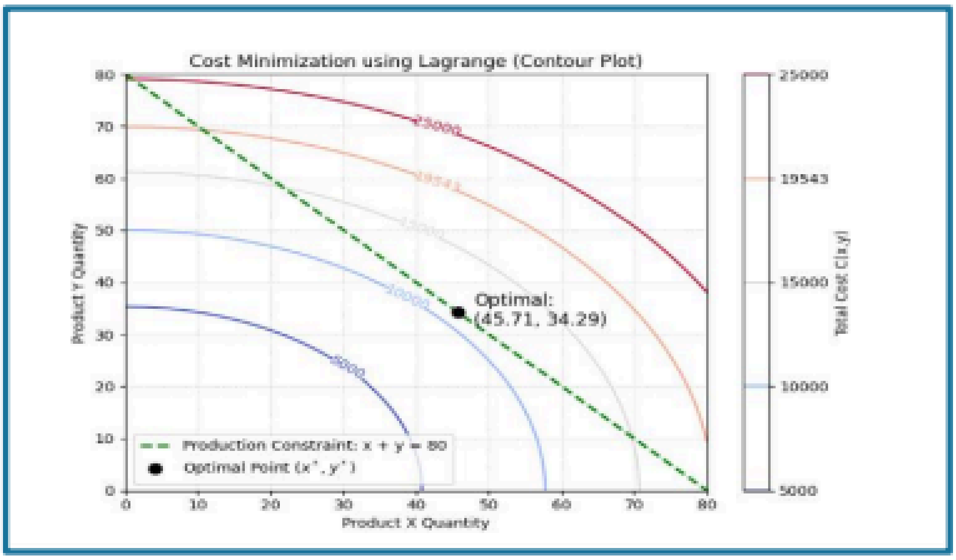


Figure 1: Cost Minimization Contour Plot

8. Results & Discussion

Interpretation of the Solution :

The analysis successfully determined the optimal production allocation to satisfy the required quota ($x+y=80$) at the minimum possible cost.

- Optimal Production: The optimal quantities are $x^* = 45.71$ units of Product X and $y^* = 34.29$ units of Product Y.
- Minimum Cost: This specific mix results in the Minimum Cost (C^*) of 10971.43 units.
- Result Verification: The numerical output from the SLSQP algorithm was $x^*=45.7143$ and $y^*=34.2857$, which perfectly matches the analytical solution derived from the Lagrange equations.

Why does the Maximum Occur at the Boundaries of the Constraint Line?

In this cost minimization problem, the objective function is $C(x,y) = 3x^2 + 4y^2$. The function exhibits **increasing marginal cost** (the squared terms), meaning the cost rises very steeply when production of one item is high.

Maximum Cost Location: The **maximum cost** occurs at the endpoints of the constraint line segment ($x+y=80$):

- If production is skewed towards the more expensive product (Y), for example, at (0, 80), the cost is $C = 3(0)^2 + 4(80)^2 = 25600$.
- This is why, in cost functions with increasing marginal costs, the maximum value is always found at the extremes of the feasible region, where allocation is least balanced.

Why is the Minimum Limit Not Zero?

The minimum cost C^* is **10971.43**, not zero. This is because:

- **The Constraint:** The constraint is an **equality quota ($x+y=80$)**. The company is *required* to produce a total of 80 units.
- **The Cost Function:** The cost function $C = 3x^2 + 4y^2$ is inherently positive for any non-zero production. Since the quota requires $x>0$ or $y>0$, the cost cannot be zero unless the quota itself was zero.

Are the Results Logical?

Yes, the results are highly logical and reflect sound economic decision-making:

1. Optimal Allocation: The company avoided the inefficient 40/40 split. The optimal solution requires producing more of Product X (45.71) and less of Product Y (34.29).

2. Cost Rationale: This shift is logical because Product Y is more expensive (its coefficient is $4y^2$, higher than $3x^2$). The company allocates more resources to the cheaper product (X) until the marginal cost of producing the last unit of X equals the marginal cost of producing the last unit of Y, which is the exact condition enforced by the Lagrange Multiplier method.

9. Conclusion

This project successfully applied Lagrange Multipliers to solve a constrained optimization problem: minimizing the cost function $C(x,y) = 3x^2 + 4y^2$ subject to the production quota $x+y=80$.

The project confirmed two crucial aspects:

- Analytical Validation: The analytical solution determined the optimal allocation to be x^* 45.71 and y^* 34.29 , resulting in a minimum cost of C^* 10971.43

- Practical Implementation: The Python numerical implementation (SLSQP algorithm) and the Contour Plot visually verified that this optimal point is indeed the point of tangency between the lowest cost contour and the production constraint line.

In conclusion, we mastered the integration of mathematical theory with computational tools to derive a logical and economically sound decision - achieving the required production quota while strictly minimizing expenses.

10. References

- Convex Analysis for Optimization: A Unified Approach — Jan Brinkhuis
- Lagrange-type Functions in Constrained Non-Convex Optimization
- Bazaraa, M. S., Sherali, H. D., & Shetty, C. M. (2013). Nonlinear Programming: Theory and Algorithms. Wiley.

Project work summary

Name	Task
Shaden	Introduction - Problem Statement - Mathematical Formulation – Visualization - Results & Discussion - Report Creator
Wajd	Mathematical Formulation - Lagrange Method - Description of the Numerical method - Visualization - Results & Discussion - Conclusion
Taif	Mathematical Formulation -Numerical Implementation - Visualization - Conclusion Results & Discussion - Conclusion